

Quantitative Data Appendices: Roman and Byzantine Empires

Infonomics Project CAS Framework Research

This appendix provides verified quantitative data for mathematical modeling of imperial complex adaptive systems. Data spans **five indicator categories** across **three phases** for each empire, with confidence assessments and source citations for academic rigor.

PART I: ROMAN EMPIRE (264 BCE – 476 CE)

1. Economic Indicators

1.1 GDP estimates (1990 International Dollars)

The scholarly consensus on Roman GDP derives primarily from **Scheidel & Friesen (2009)** using convergent expenditure, income, and comparative methodologies.

Phase	Period	Total GDP	Per Capita GDP	Source	Confidence
Rise	264-27 BCE	Not quantified	Not quantified	—	DATA GAP
Peak	27 BCE-180 CE	~50 million tons wheat equiv. (~HS 18-20 billion)	\$610-700 (1990 Int\$)	Scheidel & Friesen (2009), JRS 99:61-91	MODERATE
Peak	14 CE (Italy)	—	857(1990Int)	Maddison (2007)	MODERATE
Decline	180-476 CE	Qualitative contraction	Declining	—	LOW

Maddison Project 2023 Data (2011 PPP \$): Roman Italy at 1 CE: **\$1,407**; Roman Greece: **\$1,275**; Roman Egypt: **\$1,116**. (Substack)

Methodology note: All ancient GDP estimates are highly speculative, derived from fragmentary evidence. Scheidel & Friesen used three convergent methods: expenditure-based summation, income distribution modeling, and comparative wage/GDP ratios from better-documented pre-industrial economies.

1.2 Trade volume estimates

Metric	Value	Source	Confidence
Indian Ocean imports (1st c. CE)	~1,000 million HS/year	McLaughlin	LOW-MODERATE
Red Sea maritime tax (tetarte)	25% import duty	Documentary	HIGH
Red Sea trade tax revenue	200-270 million HS/year (upper bound)	Various estimates	LOW

1.3 Tax revenue and fiscal structure

Period	State Revenue	Tax as % GDP	Source	Confidence
Pre-Pompey Republic	200 million HS	—	Imperium Romanum	MODERATE
Post-Pompey (c. 60 BCE)	540 million HS	—	Imperium Romanum	MODERATE
Peak (~150 CE)	~900 million HS	5-7%	Duncan-Jones (1994); Hopkins (1980)	MODERATE
Scheidel model	~1 billion HS extracted	~5% of GDP	Scheidel & Friesen	MODERATE

Key finding: Roman taxation was comparatively light at **5-7% of GDP**, similar to France in 1700. Wikipedia +2
Municipal taxes were additional.

1.4 Urbanization rates

Phase	Urban %	Total Urban Pop.	Source	Confidence
Peak (1st-2nd c. CE)	20-25% empire-wide	14-15 million	Hanson & Ortman (2017), JRA	MODERATE
Peak (Italy)	25-40%	—	Wilson (2011); Scheidel	MODERATE
Decline (4th-5th c.)	10-15%	Declining	Russell (1958)	LOW

Urban site count: **1,388-2,000** identified across empire. Peak urbanization comparable to mid-18th century Europe. Historum

1.5 Agricultural output (Grain annona)

Metric	Value	Source	Confidence
Rome's annual grain requirement	150,000-300,000 tons	Rickman (1980); Erdkamp	MODERATE
Grain dole recipients	~200,000 male citizens	Augustan records	HIGH
Monthly ration per recipient	5 modii (~33-35 kg)	Documentary	HIGH
Egypt's supply share	~4 months/year	Josephus	MODERATE
Africa's supply share	~8 months/year	Josephus	MODERATE

2. Demographic Indicators

2.1 Total population by period

The "high count" (Lo Cascio) versus "low count" (Beloch/Scheidel) debate creates **40-50% variance** in estimates.

Phase	Period	Population	Source	Confidence
Rise	264 BCE	4-5 million citizens; ~14 million Italy	Brunt (1987); Scheidel (2007)	MODERATE
Peak	165 CE	70-76 million empire-wide	Harper (2017); Scheidel (2007); Frier (2000)	MODERATE
Peak	Augustus (14 CE)	~46 million rising to 61 million by 164 CE	Frier (2000), CAH	MODERATE
Decline	350 CE	39-50 million	Russell (1958); Lo Cascio (1994)	LOW

Population density at peak: ~20 persons/km² (Harper). [Wikipedia](#) [Ancient-history-sites](#)

2.2 City of Rome population

Phase	Population	Source	Confidence
Late Republic (c. 50 BCE)	500,000-800,000	Morris; grain dole analysis	MODERATE

Phase	Population	Source	Confidence
Peak (1st-2nd c. CE)	~1,000,000	Scholarly consensus	HIGH
Early Decline (300 CE)	~500,000	Russell (1958)	MODERATE
Late Decline (late 5th c.)	~50,000	Urban decline studies	MODERATE

2.3 Plague and crisis mortality

Plague	Period	Mortality Estimate	Deaths	Source	Confidence
Antonine Plague	165-180 CE	7-10% of population	7-8 million	Littman & Littman (1973); Harper (2017)	MODERATE
Antonine (Rome peak)	189 CE	2,000 deaths/day	—	Cassius Dio	MODERATE
Cyprian Plague	249-270 CE	10-25% in affected areas	5,000/day at peak (Rome)	Cyprian, De Mortalitate; Harper (2015)	LOW-MODERATE
Cyprian (Alexandria)	250s CE	>60% population loss	500,000 → 190,000	Dionysius of Alexandria	LOW

Scholarly debate: Antonine Plague estimates range from 2-33%; consensus centers on 7-10%. Kyle Harper identifies Cyprian Plague as possible viral hemorrhagic fever based on symptom descriptions. Wikipedia Disease identification remains contested.

2.4 Military manpower as % of population

Phase	Army Size	Population	% Military	Source	Confidence
Rise (Republic)	30,000-150,000	4-14 million	0.75-3.6%	Polybius	MODERATE
Peak (Tiberius)	250,000	60-70 million	0.4-0.5%	Le Bohec	MODERATE-HIGH
Peak (211 CE Severan)	450,000	70-75 million	0.6-0.75%	Hopkins	MODERATE

Phase	Army Size	Population	% Military	Source	Confidence
Decline (4th c.)	400,000- 600,000	40-50 million	1.0-1.5%	Jones; Notitia Dignitatum	MODERATE

2.5 Urban versus rural distribution

Phase	Urban	Rural	Source	Confidence
Peak	20-25%	75-80%	Hanson & Ortman (2017)	MODERATE
Italy (Peak)	25-40%	60-75%	Scheidel (2007)	MODERATE
Decline	10-15%	85-90%	Russell (1958)	LOW

3. Financial Indicators

3.1 Currency debasement: Denarius silver content

This represents one of the best-documented quantitative trends, derived from **Kevin Butcher & Matthew Ponting (2014/2015)** metallurgical analysis of 1,300+ coins.

Period/Emperor	Silver Content	Weight	Source	Confidence
Introduction (211 BCE)	95%	4.5g	Crawford (1974); NGC	HIGH
Augustus (27 BCE-14 CE)	95-98%	~3.9g	TheCollector; NGC	HIGH
Tiberius (14-37 CE)	98-99%	—	JM Bullion	HIGH
Nero (64 CE reform)	80%	Reduced	Butcher & Ponting (2015)	HIGH
Trajan (107 CE)	89-90%	—	NGC	HIGH
Marcus Aurelius	~75%	—	Hard Money History	HIGH
Commodus (192 CE)	~71%	—	NGC	HIGH
Septimius Severus	46-57%	—	NGC	HIGH
Caracalla (Antoninianus, 215 CE)	~50%	5.1g	TheCollector	HIGH
Gordian III (244 CE)	~40%	—	TheCollector	HIGH
Gallienus (268 CE)	2.5-5%	~3g	NGC; Wikipedia	HIGH
Diocletian era	<5%	—	Multiple	HIGH

Key inflection points: Nero's 64 CE reform, Septimius Severus (193 CE), and Gallienus (268 CE) mark critical debasement accelerations. Total trajectory: **95% → 5%** over ~500 years.

3.2 Interest rates

Period	Rate	Type	Source	Confidence
Twelve Tables (450 BCE)	8.33% max	Legal limit	Lacus Curtius	HIGH
342 BCE (Genucian laws)	0% (banned)	Legal	Lacus Curtius	HIGH
Late Republic-Empire	12% (1%/month)	Legal max (centesimae usurae)	EH.net; Multiple	HIGH
Augustus (25 BCE)	~4%	Market low	Armstrong Economics	MODERATE
33 CE Crisis	12%+	Market stress	Harvard Epicenter	HIGH

Period	Rate	Type	Source	Confidence
Provinces	12-48%	Provincial lending	Armstrong Economics	MODERATE
Egyptian grain loans	Up to 50%	Specialized	BTCM Research	HIGH
Pompeii pawnbrokers	45-75%	Informal sector	TIME	HIGH

Note: Romans lacked bond markets. State debt managed through currency debasement, tax increases, or direct spending rather than bond issuance.

3.3 Price indices and inflation

Period	Inflation	Source	Confidence
1st-2nd century CE	~1%/year or less	Scheidel (2009); Rathbone	MODERATE
Wheat (1st-2nd c.)	~0.5 denarii/modius	Temin; Jones	HIGH
3rd Century Crisis	~15,000% (200-300 CE)	Acton Institute	MODERATE
Wheat (1st c. → 4th c. Egypt)	6 drachmas → 2+ million drachmas	Acton Institute	HIGH

Diocletian's Edict (301 CE): Maximum wheat price set at 100 denarii communes/modius; farm laborer wage at 25 denarii/day.

3.4 State budget allocation

Category	Amount	% of Budget	Source	Confidence
Military spending (~150 CE)	~670 million HS	~75-80%	Duncan-Jones	MODERATE
Total budget (~150 CE)	~800-900 million HS	100%	Duncan-Jones	MODERATE
Military as % GDP	—	~2.5%	Duncan-Jones	MODERATE

4. Institutional Indicators

4.1 Number of rulers per phase

Phase	Period	Years	Rulers	Emperors/Decade	Source	Confidence
Rise (Republic)	264-27 BCE	237	N/A (Consular)	2 consuls/year	—	HIGH
Peak	27 BCE-180 CE	207	~17 emperors	0.8/decade	Nature (Saleh 2019)	HIGH
Decline	180-476 CE	296	~50+ emperors	1.7/decade	Royal Society (2021)	HIGH
Crisis (3rd c.)	235-284 CE	49	26-80 claimants	5+/decade	—	HIGH

4.2 Violent versus peaceful succession rates

Study	Sample	Violent Death Rate	Natural Death Rate	Source	Confidence
Saleh (2019)	69 emperors (27 BCE-395 CE)	62%	38%	Nature	HIGH
Royal Society (2021)	175 emperors (incl. Byzantine)	75%	25%	Royal Society Open Science	HIGH

Breakdown (Miller analysis): Assassinated: 48%; Died in battle: 8%; Executed/other violent: ~12%.

4.3 Average reign duration

Condition	Average Reign	Source	Confidence
Natural death	~12 years	Royal Society (2021)	HIGH
Violent death	~5 years	Royal Society (2021)	HIGH
Overall average	~8 years	Derived	HIGH
Hazard rate change point	13 years	Power-law analysis	MODERATE

Hereditary succession correlated with significantly longer reigns than usurpation ($p < 0.001$).

4.4 Major legal codifications

Codification	Date	Phase	Significance
Twelve Tables	451-450 BCE	Pre-Rise	Foundation of Roman law
Lex Aebutia	~150 BCE	Rise	Reformed legal procedure
Codex Gregorianus	~291 CE	Decline	First systematic imperial constitution collection
Codex Hermogenianus	~295 CE	Decline	Supplement to Gregorianus
Codex Theodosianus	438 CE	Decline	Theodosius II's compilation

Count: 5 major codifications during Roman period.

4.5 Civil wars per century

Period	Civil Wars	Rate	Source	Confidence
Rise (2nd-1st c. BCE)	5-8 major	~3-4/century	Wikipedia List	HIGH
Peak (1st-2nd c. CE)	2-3 (Year of Four Emperors)	~1/century	—	HIGH
3rd Century Crisis	26+ in 50 years	~52/century	—	HIGH
4th century	5-7 major	~6/century	—	MODERATE
5th century	Multiple usurpations	Continuous	—	MODERATE

Note: Only two decades between 190-480 CE passed without Roman civil war (OUP Blog 2022).

5. Military Indicators

5.1 Army size (peak strength)

Phase	Period	Legions	Total Army	Source	Confidence
Rise (Punic Wars	216 BCE	23	~150,000	Polybius	MODERATE

Phase	Period	Legions	Total Army	Source	Confidence
peak)					
Rise (Late Republic)	44 BCE	37-60	100,000-150,000	—	MODERATE
Peak (Augustus)	End of reign	25	~300,000	Le Bohec	HIGH
Peak (Severan)	211 CE	33	~450,000	Hopkins; Le Bohec	MODERATE
Decline (4th c.)	Diocletian/Constantine	64-67	400,000-600,000	Jones; Notitia Dignitatum	MODERATE

Peak breakdown (Tiberius era): Legionaries: 125,000; Auxiliaries: 125,000; Praetorians: 9,000; Urban Cohorts: 3,000; Marines: 40,000.

5.2 Navy size

Phase	Period	Fleet Size	Source	Confidence
Rise (First Punic War)	256 BCE	330 warships	Polybius	HIGH
Rise (Actium)	31 BCE	~500 warships	Historical accounts	MODERATE
Peak	1st-2nd c. CE	200-300 warships	—	MODERATE

Fleet personnel: 30,000-40,000 total (Mediterranean fleets: 15,000-20,000). [Wikipedia](#)

5.3 Defense spending as % of revenue

Period	Military Cost	% of Budget	% of GDP	Source	Confidence
Peak (~150 CE)	~670 million HS	~80%	~2.5%	Duncan-Jones	MODERATE
Decline (4th c.)	~63,250 lbs gold	~21% of budget	Higher	Elton estimates	MODERATE

5.4 Major battles won/lost ratio

Phase	Approximate Ratio	Notable Examples	Confidence
Rise	~10:1	Zama (202 BCE), Pydna (168 BCE)	LOW

Phase	Approximate Ratio	Notable Examples	Confidence
Peak	~15:1	Dacian Wars victories	LOW
Decline	~2:1 to 1:1	Adrianople (378 CE - defeat), Teutoburg (9 CE - defeat)	LOW

Major defeats: Teutoburg Forest (9 CE): 3 legions lost; Carrhae (53 BCE): 20,000 killed; Adrianople (378 CE): Emperor Valens killed, ~20,000 dead.

5.5 Frontier length and garrison requirements

Frontier Section	Length (km)	Features	Source	Confidence
Total Roman Limes	5,000-7,500 km	Atlantic to Black Sea to Red Sea	UNESCO World Heritage	HIGH
Upper Germanic-Raetian	550 km	60+ forts, 900 watchtowers	UNESCO	HIGH
Hadrian's Wall	118 km	Stone wall 6m tall	UNESCO	HIGH
Danube Limes (Western)	600 km	River border	UNESCO	HIGH
Limes Arabicus	1,500 km	Eastern desert frontier	Wikipedia	MODERATE
Fossatum Africae	750+ km	African ditch system	UNESCO	MODERATE

Garrison requirement: ~300,000-400,000 troops for frontier defense.

PART II: BYZANTINE EMPIRE (330 – 1453 CE)

1. Economic Indicators

1.1 GDP estimates (1990 International Dollars)

Byzantine GDP quantification derives primarily from **Branko Milanovic (2006)** and **Warren Treadgold (1997)**.

Phase	Period	Per Capita GDP	Total GDP (Implied)	Source	Confidence
Rise	330-565 CE	Not quantified	—	—	DATA GAP
Peak	~1000 CE (Basil II)	\$680-770 (1990 Int\$) = ~6 nomismata/year	\$8-14 billion (calculated)	Milanovic (2006), Review of Income & Wealth	MODERATE
Decline	1025-1453 CE	Declining to ~2N by 1453	—	Lambert (2022)	LOW

Comparative finding: Byzantine per capita income at its 1000 CE peak was approximately **20% below** Roman levels under Augustus. (SSRN) (Cuny)

Methodology note: Milanovic derived estimates from Treadgold's Byzantine state budget data, assuming budgets = 5-8% of GDP, then converting via subsistence basket comparisons. (ResearchGate)

1.2 Tax revenue and fiscal structure

Period	State Revenue	Tax as % GDP	Source	Confidence
Peak (~1025 CE)	5.9 million nomismata	5-8%	Treadgold; Morrisson (2002)	MODERATE
840-867 CE	20,000 lbs gold	—	Treadgold	MODERATE
Land tax share	—	80-95% of revenue	Hendy (1985)	MODERATE
Average tax rates	~10-12% of land values	—	Various	MODERATE

1.3 Urbanization rates

Metric	Estimate	Source	Confidence
Rural population	~90%	Milanovic (2006); Morrisson & Cheynet	HIGH
Urban population	~10%	Milanovic (2006)	HIGH
Constantinople share of total	~3% at peak	Milanovic (2006)	HIGH
Byzantine vs Europe (c. 1000)	Higher due to Constantinople	Milanovic	MODERATE

2. Demographic Indicators

2.1 Total population by period

Population figures derive primarily from **Warren Treadgold's** fiscal reconstruction methodology.

Phase	Year	Population	Source	Confidence
Rise	330 CE	~18 million (Eastern Empire)	Treadgold (1997)	MODERATE
Rise	540 CE (pre-plague)	26-27 million	Treadgold (1997)	MODERATE
Rise	560 CE (post-plague)	19.5 million	Treadgold (1997), p.59	MODERATE
Peak	641 CE (post-Arab losses)	10.5 million	Treadgold (1997)	MODERATE
Peak	774 CE	7 million	Treadgold (1995), p.67	MODERATE
Peak	1025 CE (Basil II)	12-18 million	Treadgold (2001); Harl (1996)	MODERATE
Decline	1204 CE (pre-Fourth Crusade)	~9 million	Byzantine-Empire.com	LOW
Decline	1282 CE	3-5 million	Treadgold	MODERATE
Decline	1312 CE	~2 million	Treadgold (1997, 2020)	MODERATE
Decline	1453 CE	~500,000	Laiou & Morrisson (2007)	MODERATE

2.2 Constantinople population

Phase	Year	Population	Source	Confidence
Rise	4th-5th c.	250,000-500,000	Metropolitan Museum	MODERATE
Rise	542 CE (pre-plague)	~500,000	Magdalino; Milanovic	MODERATE
Peak	7th c. (nadir)	~70,000	Magdalino; Ousterhout	LOW
Peak	10th-12th c.	300,000-400,000	Harvey (2003), p.307; Villehardouin	MODERATE

Phase	Year	Population	Source	Confidence
Decline	1204 CE (post-Crusade)	~30,000	Kaldellis	LOW
Decline	1453 CE	25,000-50,000	Treadgold (1997)	HIGH

2.3 Plague mortality

Justinianic Plague (541-542 CE and recurrences)

Metric	Estimate	Source	Confidence
Constantinople daily deaths (peak)	5,000-10,000/day	Procopius (5,000 modern; 10,000 ancient)	MODERATE
Constantinople total mortality	40% (~200,000-300,000)	Traditional; John of Ephesus	MODERATE
Empire-wide initial mortality	20-25% (traditional)	Maximalist view	LOW-MODERATE
Total deaths (541-750 CE)	25-100 million	Various sources	LOW
Recurrences	18-21 outbreaks over 200 years	Stathakopoulos	HIGH
Average interval	7.7 years between outbreaks	—	MODERATE

CRITICAL REVISIONIST VIEW: Mordechai & Eisenberg (2019, PNAS) argue plague mortality was "far lower than previously believed" and did not cause widespread demographic decline. They state: "We have no more basis for the current maximalist estimate of a 50% demographic loss than we would have for a minimalist estimate of 0.1%." This remains an **active scholarly debate**.

Black Death in Byzantium (1347-1453)

Metric	Data	Source	Confidence
First arrival	1347 CE	Multiple	HIGH
Total plague reports (1347-1453)	61 documented outbreaks	Tsiamis et al. (2011)	HIGH
Major epidemic waves	9 waves	Tsiamis et al. (2011)	HIGH
Average wave duration	3.2 years	Tsiamis et al. (2011)	MODERATE
Byzantine-specific mortality	NOT QUANTIFIED	—	DATA GAP

2.4 Military manpower as % of population

Period	Army Size	Population	% Military	Source	Confidence
559 CE	150,000	~20 million	0.75%	Treadgold	MODERATE
773 CE	80,000	7 million	1.14%	Arab sources; Haldon	MODERATE
840 CE	120,000-154,600	8 million	1.5-1.9%	Treadgold; Arab geographers	MODERATE
1025 CE	200,000-250,000	12-18 million	1.4-1.7%	Treadgold (1997)	LOW-MODERATE

Scholarly debate: Treadgold's estimates are considered high by John Haldon. Conservative revisions reduce these by 30-50%.

2.5 Urban versus rural distribution

Metric	Estimate	Source	Confidence
Rural	~90%	Milanovic; Morrisson & Cheynet	HIGH
Urban	~10%	Milanovic	HIGH

3. Financial Indicators

3.1 Currency stability: Solidus/Nomisma gold content

The Byzantine solidus represents **one of history's most stable currencies**, maintaining purity for

approximately 700 years.

Golden Age of Stability (312-1034 CE)

Metric	Value	Source	Confidence
Constantine I introduction (312 CE)	4.5g at 24 carats (~99% gold)	Wikipedia; World History Encyclopedia	HIGH
Standard (312-1030s CE)	4.45-4.5g at 95.5-98% gold	Wikipedia; Grierson (1982)	HIGH
Duration of stability	~700 years	Philip Grierson, Byzantine Coinage	HIGH

Debasement Timeline (1034-1092 CE)

Emperor	Reign	Gold Content	Source	Confidence
Pre-debasement	to 1034	95.5-98% (23+ carats)	Wikipedia; MDPI	HIGH
Michael IV	1034-1041	Debasement begins	Wikipedia	HIGH
Constantine IX	1042-1055	~87.5% (21 carats)	Wikipedia; MDPI	HIGH
Constantine X	1059-1067	~75% (18 carats)	Wikipedia	HIGH
Romanus IV	1068-1071	~66.7% (16 carats)	Wikipedia	HIGH
Michael VII	1071-1078	~58% (14 carats)	Wikipedia	HIGH
Nicephorus III	1078-1081	~33% (8 carats)	Wikipedia	HIGH
Early Alexios I	1081-1092	0-33% (0-8 carats)	Wikipedia	HIGH

Critical contrast with Rome: Byzantine debasement was rapid once started—from 98% to nearly 0% gold in ~50 years (1034-1081)—compared to Rome's gradual 500-year silver decline.

Hyperpyron Era (1092-1453 CE)

Period	Gold Content	Source	Confidence
Alexios I reform (1092)	85.4% (20.5 carats), 4.45g	Wikipedia; Grokipedia	HIGH
Post-1204 (Latin Empire)	18-20 carats	Grokipedia	MODERATE
Mid-14th century	~50% fineness	Wikipedia	MODERATE
Final hyperpyra (1347-1352)	Last gold coins struck	Wikipedia	HIGH

3.2 Interest rates (Justinian's Code, 533 CE)

Byzantine interest rates were **lower and more regulated** than Roman rates, with a sliding scale by creditor type.

Creditor Type	Maximum Rate	Source	Confidence
Senators/"Illustrious persons"	4%	EH.net; BTCM	HIGH
Farmers	4%	EH.net	HIGH
Regular citizens (non-business)	6%	EH.net; BTCM	HIGH
Business loans	8%	EH.net; BTCM	HIGH
Bankers	8%	BTCM	HIGH
Maritime loans (foenus nauticum)	12%	EH.net; BTCM	HIGH
Churches/foundations	3% maximum	BTCM	MODERATE

Anti-compound interest: Accumulated interest could not exceed principal amount.

Century	Banker Rate	Source	Confidence
6th century	8%	BTCM	HIGH
9th-10th century	11.125%	BTCM; PitchBook	MODERATE

4. Institutional Indicators

4.1 Number of emperors per phase

Phase	Period	Duration	# Emperors	Avg Reign	Source	Confidence
Rise	330-565 CE	235 years	~20	~12 years	List of Byzantine Emperors	HIGH
Peak	565-1025 CE	460 years	~45	~10 years	Wikipedia	HIGH
Decline	1025-1453 CE	428 years	~29	~15 years	Wikipedia	HIGH
TOTAL	330-1453 CE	1,123 years	94 emperors	~12 years	Wikipedia	HIGH

4.2 Violent versus peaceful succession rates

Metric	Data	Source	Confidence
Emperors who rose as usurpers	20 (21%)	Wikipedia	HIGH
Violent ends/depositions	>30 emperors	Wikipedia	MODERATE
Violent succession estimate	30-40%	Derived	MODERATE

Contrast with Rome: Byzantine violent succession rate (30-40%) was significantly lower than Roman rate (62-75%).

4.3 Major legal codifications

Codification	Date	Description	Source	Confidence
Corpus Juris Civilis	529-534 CE	Justinian's comprehensive code: Codex, Digest, Institutes, Novellae	Standard	HIGH
Ecloga	726 CE	Leo III's legal revision	Standard	HIGH
Basilika	886-912 CE	Basil I & Leo VI's Greek translation (60 books)	Standard	HIGH
Hexabiblos	1345 CE	Constantine Harmenopoulos's manual	Standard	HIGH

Count: 4 major codifications.

4.4 Civil wars and usurpations by century

Century	Major Conflicts	Notable Events
5th (400-500)	Few	Basiliscus usurpation (475-476)
6th (500-600)	Minimal	Nika Revolt (532)
7th (600-700)	Multiple	Twenty Years' Anarchy (695-717)
8th (700-800)	Multiple	Iconoclast conflicts
9th (800-900)	Moderate	Michael II (820); Basil I (867)
10th (900-1000)	Stable	Macedonian dynasty peak
11th (1000-1100)	Multiple	Post-Basil II crisis (1025-1081)
14th (1300-1400)	Severe	Civil wars 1320s, 1340s

5. Military Indicators

5.1 Army size (Tagmata and Themata system)

Period	Total Army	Tagmata	Thematic	Marines/Naval	Source	Confidence
559 CE	150,000	—	—	—	Treadgold	MODERATE
773 CE	80,000	—	—	—	Arab sources	MODERATE
774 CE	118,400	18,000	62,000	38,400	Treadgold (1995)	MODERATE
840 CE	120,000-154,600	24,000	96,000	34,200	Treadgold; Arab geographers	MODERATE
959 CE	148,000	—	—	—	Treadgold	MODERATE
1025 CE	200,000-250,000	—	—	—	Treadgold (1997)	LOW

Scholarly debate: Treadgold estimates tagmata at 4,000 men each (24,000 total); Haldon estimates 1,000-1,500 men each (~4,000-5,000 total).

5.2 Navy size (Dromon fleet strength)

Period	Fleet Size	Personnel	Source	Confidence
717-718 CE	Effective defense force	—	Defeated Arab fleet of ~1,800 ships	HIGH
9th-10th century baseline	~240 warships	—	Treadgold	MODERATE
960-961 CE	307 warships	—	Treadgold	MODERATE
Mid-9th century peak	~500 ships	—	Grokipedia	LOW

Ship crews: Heavy dromon: 230 rowers + 70 marines (300 total); Ousiakon: 108 men.

5.3 Defense spending as % of revenue

Metric	Estimate	Source	Confidence
Military budget (840-867 CE)	20,000 lbs gold	Treadgold	MODERATE
Roman comparison	60-80% of budget	Economics of Roman Army	MODERATE
Byzantine estimate	~50-80% of budget	Derived	LOW

5.4 Major battles won/lost ratio by phase

Phase	Assessment	Notable Events
Rise (330-565)	Net positive	Vandalic War (533), Gothic War victories
Peak (565-1025)	Mixed → Positive	Constantinople defense (717-718); Basil II's Bulgarian victories; but loss of Syria/Egypt
Decline (1025-1453)	Net negative	Manzikert (1071); Fourth Crusade (1204); Ottoman defeats

Data gap: Precise win/loss statistics not compiled in available sources.

5.5 Frontier length and garrison

Data gap: Specific frontier lengths not documented with precision for Byzantine Empire. General observations:

- Pre-Arab conquest: Empire stretched from Spain to Mesopotamia

- Post-7th century: Reduced to Anatolia + Balkans core
- Theme system distributed defense across **20+ themes** by 840 CE
- Each theme garrisoned **4,000-10,000 troops**

PART III: Cross-Empire Comparison

Key quantitative comparisons

Metric	Roman Empire (Peak)	Byzantine Empire (Peak)	Notes
Per capita GDP	\$610-700 (1990 Int\$)	\$680-770 (1990 Int\$)	Byzantine ~20% lower than Augustan Rome
Peak population	70-76 million	12-18 million (1025 CE)	Roman 4-6x larger
Capital population	~1 million (Rome)	300,000-400,000 (Constantinople)	Rome 2.5-3x larger
Currency stability	500 years of debasement (95%→5%)	700 years of stability (95-98%), then rapid collapse	Byzantine stability superior
Violent succession rate	62-75%	30-40%	Byzantine more stable
Military as % population	0.4-0.75%	1.14-1.9%	Byzantine higher extraction rate
Interest rate (legal max)	12% (centesimae usurae)	4-12% (sliding scale)	Byzantine more regulated
Total emperors	~69 (27 BCE-395 CE)	94 (330-1453 CE)	Byzantine longer duration
Average reign	~8 years	~12 years	Byzantine reigns longer

Plague mortality comparison

Plague	Empire	Mortality Estimate	Confidence
Antonine (165-180 CE)	Roman	7-10%	MODERATE (consensus)
Cyprian (249-270 CE)	Roman	10-25%	LOW-MODERATE

Plague	Empire	Mortality Estimate	Confidence
Justinianic (541-542 CE)	Byzantine	20-50% (contested)	LOW-MODERATE
Black Death (1347-1453)	Byzantine	Not quantified	DATA GAP

Data Gaps and Limitations

The following data **cannot be found** or remain **highly contested**:

1. **Roman Rise-period GDP:** No quantitative estimates exist for 264-27 BCE.
2. **Byzantine Rise-period GDP:** Pre-Justinianic economic quantification unavailable.
3. **Decline-phase GDP trajectories:** Only qualitative contraction documented; no numerical series.
4. **Black Death Byzantine-specific mortality:** No quantified data despite 61 documented outbreaks.
5. **Precise battle won/lost ratios:** Not compiled systematically for either empire.
6. **Byzantine frontier lengths:** General estimates only; no UNESCO-level documentation like Roman limes.
7. **Mint output volumes:** Fragmentary for both empires.
8. **Agricultural output beyond grain:** Limited to annona data for major cities.
9. **Justinianic Plague true mortality:** Active scholarly debate between maximalists (25-50%) and revisionists (minimal impact).

Confidence Assessment Summary

Category	Roman Empire	Byzantine Empire
GDP per capita estimates	MODERATE	MODERATE
Total population	MODERATE (40-50% variance)	MODERATE
Plague mortality	MODERATE (Antonine); LOW (Cyprian)	LOW-MODERATE (Justinianic); DATA GAP (Black Death)
Currency debasement	HIGH (numismatic analysis)	HIGH (numismatic analysis)

Category	Roman Empire	Byzantine Empire
Interest rates	HIGH	HIGH
Military size	MODERATE-HIGH	MODERATE (Treadgold vs Haldon debate)
Institutional metrics	HIGH	HIGH
Fiscal/budget data	MODERATE	MODERATE
Frontier/garrison	HIGH (UNESCO)	LOW (fragmentary)

Key Scholarly Sources

Roman Economy:

- Scheidel & Friesen (2009), "Size of the Economy," *JRS* 99:61-91
- Peter Temin (2013), *The Roman Market Economy* (Princeton)
- Duncan-Jones (1994), *Money and Government in the Roman Empire*
- Hopkins (1980), "Taxes and Trade," *JRS* 70:101-125

Byzantine Economy:

- Milanovic (2006), *Review of Income and Wealth* 52(3):449-470
- Treadgold (1995, 1997), *Byzantium and Its Army; History of Byzantine State*
- Laiou & Morrisson (2007), *The Byzantine Economy* (Cambridge)
- Hendy (1985), *Studies in the Byzantine Monetary Economy*

Numismatics:

- Butcher & Ponting (2014/2015), *Metallurgy of Roman Silver Coinage* (Cambridge)
- Grierson (1982/1999), *Byzantine Coinage* (Dumbarton Oaks)
- Crawford (1974), *Roman Republican Coinage*

Demographics & Plague:

- Harper (2017), *The Fate of Rome* (Princeton)
- Littman & Littman (1973), "Galen and the Antonine Plague," *AJP* 94:243-255

- Mordechai & Eisenberg (2019), PNAS - revisionist Justinianic Plague
- Stathakopoulos (2004), *Famine and Pestilence* (Birmingham)
- Little, ed. (2007), *Plague and the End of Antiquity*

Institutional:

- Saleh (2019), *Nature* - statistical analysis of emperor deaths
- Royal Society Open Science (2021) - power-law analysis

Maddison Project:

- University of Groningen Maddison Project Database (2020/2023)
- Maddison (2001/2007), *The World Economy: A Millennial Perspective*